GEK-121341B

ActivePoint™ HMI Philosophy
Design Guide

GE VERNOVA

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Document Updates

Location	Description
Key Performance Indicators (KPIs)	New section
Throughout the document	GE Vernova branding

Acronyms and Abbreviations

BOP	Balance of Plant
GT	Gas Turbine
HMI	Human-machine Interface
HRSG	Heat Recovery Steam Generator
KPI	Key Performance Indicator
P&ID	Process and Instrumentation Diagram
ST	Steam Turbine
UX	User Experience

Related Documents

Document #	Title
GEK-121342	ActivePoint HMI Design Guide
GEK-121343	ActivePoint HMI Alarm Design Guide
GEK-121344	ActivePoint HMI Object Design Guide

Safety Symbol Legend



Warning

Indicates a procedure or condition that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure or condition that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure or condition that should be strictly followed to improve these applications.

Control System Warnings



Warning

To prevent personal injury or damage to equipment, follow all equipment safety procedures, Lockout Tagout (LOTO), and site safety procedures as indicated by Employee Health and Safety (EHS) guidelines.



Warning

This equipment contains a potential hazard of electric shock, burn, or death. Only personnel who are adequately trained and thoroughly familiar with the equipment and the instructions should install, operate, or maintain this equipment.



Warning

Isolation of test equipment from the equipment under test presents potential electrical hazards. If the test equipment cannot be grounded to the equipment under test, the test equipment's case must be shielded to prevent contact by personnel.

To minimize hazard of electrical shock or burn, approved grounding practices and procedures must be strictly followed.



Warning

To prevent personal injury or equipment damage caused by equipment malfunction, only adequately trained personnel should modify any programmable machine.



Warning

Always ensure that applicable standards and regulations are followed and only properly certified equipment is used as a critical component of a safety system. Never assume that the Human-machine Interface (HMI) or the operator will close a safety critical control loop.

Contents

1 Introduction 9
1.1 Intended Use 9
1.2 Scope 9
1.2.1 Product Perspective 9
1.2.2 User Community and Characteristics 9
1.3 Research Background 10
2 User Experience Philosophy 11
2.1 UX Premises for HMI Design 11
2.2 Goal of HMI UX 11
2.3 UX Principles for HMI Design 11
2.4 Overall Design Principles 12
2.5 Key Performance Indicators (KPIs) 15
2.5.1 Recommended Guidelines 15
3 HMI Screen Organization and Design 17
3.1 System vs. Process Definition 17
3.2 General Screen Template 18
3.3 General Faceplate Template 20
3.4 Level Based Screen Hierarchy 22
3.5 Level Based Interaction Design Guidelines 23
3.5.1 Level 1 Screens 24
3.5.2 Level 2 Screens 25
3.5.3 Level 3 Screens 27
3.5.4 Level 4 Screens 28
3.6 Task Based Screens 29
4 Alarm Management Philosophy 31
5 Alarm Rationalization 33
5.1 Process 34
5.2 Definition Phase 35
5.3 Categorization Phase 37
5.4 Prioritization Phase 38

1 Introduction

1.1 Intended Use

This design guide documents the philosophy for the ActivePoint™ Human-Machine Interface (HMI) of GE Vernova, comprising user experience and alarm rationalization. This HMI design philosophy represents the best-in-class aspects as defined by research conducted by GE Vernova and industry standards.

The goal of the user experience and alarm rationalization is to improve Operator's awareness of plant conditions and to allow for faster response when actions are required. The HMI screens of all GE Vernova power generation equipment conform to this design guide to provide a common look-and-feel across the entire plant and to create an identity that is both visually appealing and unique to GE Vernova.

This philosophy has been established in a way that allows additions as appropriate when new situations arise. However, the fundamental user experience characteristics and alarm rationalization approach specified within this guide shall be continuously adhered to.

1.2 Scope

1.2.1 Product Perspective

The scope of this document is to introduce customers and partners to the design philosophy behind the standards. This document provides guidance for developing and modifying HMI screens in an effort to establish User Experience (UX) compliant Operator HMI screens across GE Vernova. This guide includes definition of the following.

- Screen organization philosophy
- Screen navigation methodology
- Screen content philosophy
- Task-orientated screen content objective
- Alarm visualization within HMI screens

For more specific details of the implementation, refer to the *ActivePoint HMI Design Guide* (GEK–121342), *ActivePoint HMI Object Design Guide* (GEK–121344) and *ActivePoint HMI Alarm Design Guide* (GEK–121343).

1.2.2 User Community and Characteristics

The user community for this document is:

- Customers and clients
- Partners

1.3 Research Background

This guide has been developed for the GE Vernova Mark* VIe control system for Electrical, Generators, Gas Turbines, Steam Turbines, and Plant controls.

The team, along with external consultants, researched available industry guidelines for plant operators, ranging from process control to nuclear power generation domains. The following list summarizes the guidelines reviewed.

- EEMUA 201 - Process plant control desks utilizing human-computer interfaces: A guide to design, operational and human-computer interface issues
- ISA-TR77.60.04-1996 (R2008) Fossil Fuel Power Plant Human-Machine Interface - Electronic Screen Displays
- ISA-101 – Human machine interfaces for process automation systems
- Effective Operator Display Design: ASM Consortium Guidelines
- Effective Console Operator HMI Design (second edition): ASM Consortium Guidelines
- The High Performance HMI Handbook (PAS)
- ISO 11064, Part 5: Ergonomic design of control centers – Part 5: Displays and controls
- NUREG-0700 - Human-System Interface Design Review Guidelines
- IEEE-1023 - IEEE Recommended Practice for the Application of Human Factors Engineering to Systems, Equipment, and Facilities of Nuclear Power Generating Stations and Other Nuclear Facilities
- ISA 18.2 Alarm Management Standard
- Effective Alarm Management Practices: ASM Consortium Guidelines
- The Alarm Management Handbook (PAS)

2 User Experience Philosophy

This section describes the overall design philosophy, including UX premises and other research findings.

2.1 UX Premises for HMI Design

Personnel and equipment safety are paramount; therefore, the ability for operators to detect abnormal situations and take appropriate action is critical.

Screen design must address human factors considerations, such as the HMI includes safety-critical and time-critical operations.

HMI includes an ecosystem of screens that users typically spend eight to twelve hours per shift in front of monitoring plant conditions.

HMI involves anticipating and preventing, monitoring and detecting, and troubleshooting any issues with the operation.

2.2 Goal of HMI UX

Given the premises discussed in *UX Premises for HMI Design*, the HMI UX shall enable operator competency.

With a glance, Operators should be able to recognize which information they need to attend and what it indicates.

Furthermore, the HMI provides the following benefits to Operators.

- **Simplicity** - Improves situational awareness and decreases costly errors/lost production.
- **Intuitiveness** - One-button plant startup/shutdown decreases $1MM+ misses. (Customer estimates of lost revenue from missed remote startups and shutdowns per occurrence.)
- **Responsiveness** - Improves due to an 80% actionable alarms reduction.

2.3 UX Principles for HMI Design

Based on research insights, the following UX principles are identified.

- **Simple** - Operators recognize and understand information with ease and speed.
- **Intuitive** - Operators anticipate and recognize issues quickly and respond instantly.
- **Seamless** - Operators complete their tasks and navigate seamlessly.
- **Confident** - Operators have confidence in the controls system.

2.4 Overall Design Principles

The overall HMI design philosophy is implemented to achieve screens that meet the UX Principles discussed above. High-level concepts guide the choices we make about organization, rationalization, and construction of interactive user interfaces. Described here are the nine Design Principles that guide and underpin the decisions and rationale of the ActivePoint HMI Design Guidelines. Additionally, these Design Principles inform decisions on future design directions.

Simple

1. Minimalism

To support operator understanding and situational awareness, utilizing graphical representations of plant assets can be beneficial. However, unnecessary details of the graphics are avoided to reduce visual noise and clutter.

Present only the details required to differentiate assets and place data in context.

Minimize less important assets and highlight important ones.

2. Color palette

A color palette is defined for the HMI to reduce operator's cognitive workload and, at the same time, to make abnormal conditions quickly obvious. The defined color palette is used consistently throughout all screens included within the HMI. The key tradeoff is the balance of operator support using color to differentiate types of data vs. operator cognitive overload due to too many colors to decipher the implications, increased visual clutter, and noise.

Minimize the color use to reduce operator cognitive workload.

Use dull, low-contrast, and low-saturation colors for static contents and backgrounds.

Use high-contrast and high-saturation colors (with respect to chosen background colors) for abnormal state data to draw the operator's attention to the screen.

Intuitive

3. Visual coding

Visual coding refers to any visual treatment to indicate or imply certain context around data. The visual coding is used to represent critical context for data to support an operator's ability to recognize, read, and understand. Visual coding and animation are minimized and used consistently throughout all screens included within the HMI. The key tradeoff is the balance of operator support vs. operator cognitive overload due to too many different types of codings to decipher the implications, increased visual clutter, and noise.

Minimize the use of visual coding to reduce operator cognitive workload.

Do not use the same color for different implications.

Apply enough contrast to distinguish different codings.

An animated screen element is used to support operator understanding of the process. The key tradeoff is the balance of operator support vs. operator cognitive overload, increased visual clutter, and noise.

Minimize the use of animation. Use animation only when the situation requires instant operator attention.

4. Information layout

As operators form their mental models of the operation (processes and assets) based on the location of the physical equipment and assets within the plants, utilizing the P&ID for process overview to a certain extent could be useful. However, one must be careful when using P&ID directly within screen designs. Therefore, careful design decisions have to precede applying P&ID representations.

Use P&IDs to show relative relationships between equipment in an abstracted level.

Use P&IDs only to support operator understanding.

When using P&ID, the layout is consistent with a process flow, which is directed from left to right and top to bottom.

Seamless

5. Contextual data

Data can transform into useful information when provided with a proper context. Representing the context of the data (for example, targets, operating ranges, alarm limits, and process relationships) supports faster operator understanding and situational awareness.

Where possible, use normalized and relative data representations with respect to limits and ranges.

6. Trender

For control room operators, launching separate trend windows to quickly confirm process status is considered time-consuming. Trender in screen can be configured by using a pop-up that calls for pre-configured ToolboxST* Trender.

Embedded trender usage in the screen is discouraged as it can affect screen performance.

Use trends only to better support operator understanding and to reduce cognitive workload. For instance, the rate of change of data can be best represented via trending.

7. Screen hierarchy

Currently, an HMI set can comprise 50 to 200 screens for a single-block and single-shaft power plant. Therefore, categorizing the screens systematically can improve navigation efficiency and, moreover, operation efficiency. An HMI can be organized by plant topology in which the lower level of screens progressively describes more details of a process and/or assets. Also, it can be organized by process or operator task.

By plant topology:

- Level 1: Overall Plant - used for wall display and does NOT have any navigation
- Level 2: Major units (GT, ST, HRSG, BOP, Electrical, and Generator)
- Level 3: Major systems/sub-systems of the respective units
- Level 4: Sub-systems/individual equipment or devices
- Faceplate: Individual data points

By process control or operator task:

- Screens that provide a single representation of controlling a process that includes multiple systems.
- Screens that support operator tasks (startup, shut down, and so on).

Note The distinction between Levels 3 and 4 are not strict. Depending on the complexity of a major process, Level 4 screens can be placed at Level 3. Further details are discussed in *Level Based Screen Hierarchy*.

Confident

8. Navigation

Having multiple forms of navigation improves an operator's speed of screen transition and awareness of screen locations.

Screen design techniques can support navigation. Spatial dedication on screens promotes consistency, which allows users to navigate quickly through their muscle memory. Overview of a process can be utilized as the orientation to spawn off to different screens. Use shortcuts that make sense.

For intra-screen navigation, use pop-up menus for status checks or quick control changes.

In multiple menu situations, minimize movement across the screen. Apply the unofficial rule of "3." Can I get anywhere within 3 or fewer steps?

The ability to quickly step through the operator's user scenarios needs to be considered with the screen and object design. For example, responding to an alarm condition and returning the plant to a stable condition from an upset condition. The design needs to incorporate minimal steps to understand the issue and to take steps to resolve it.

9. Alarm presentation

Alarm presentation is an integral part of the HMI user experience to enhance operator sensitivity to abnormal conditions and to improve identification and response effectiveness.

Use shape, color, and number to indicate and distinguish different priorities of alarms.

Use visual effects on alarm indicators to differentiate the statuses of alarms (acknowledged, active, normal, and so on).

Implement seamless navigation mechanisms between the alarm window and the alarm indicators on HMI screens.



Attention

Do not use colors specified for alarms on other design components.

2.5 Key Performance Indicators (KPIs)

HMIs are essential to access the GE Vernova controls infrastructure throughout the plant lifetime from commissioning to commercial operation. Each phase of a plant lifecycle uses the same HMIs; however, in very different ways. The usage duty cycle of ToolboxST, Trenders, Alarm Viewers and Operator screens varies with where the site is in the plant lifecycle. Plant commissioning dynamically changes and leverages more tools than when the plant is commercial with static configuration and focused on operations. HMIs are designed with the following KPIs based on the recommended usage. For best performance, we recommend following the guidelines below; however, HMI performance can vary in cases where site warrants different operation.

First time opening or viewing previous screen that was opened.

- < 2 seconds to open
- < 1 second to update
- <= 1 second to refresh

2.5.1 Recommended Guidelines

2.5.1.1 HMI Usage

- Three CIMPLICITY screens
- One Alarm Viewer (using engineered views and filters)
- Two Trenders
- One ToolboxST with two devices opened

2.5.1.2 CIMPLICITY Screens

There is a direct correlation of screen performance and the number of screens open.

- Limit an HMI to three concurrent CIMPLICITY screens (excluding pop-ups). This balances useful content with graphics performance. In some phases of the plant lifecycle more concurrent screens are deemed necessary.
- Close a screen not in use as opposed to minimizing the screen.

2.5.1.3 Alarm Viewer

- Limit the HMI to one Alarm Viewer using saved views provided by engineering. These views are optimized for the user experience with appropriate font sizing, pre-configured filters, and can significantly reduce CPU consumption.

2.5.1.4 Trender

Unattended trends running at controller frame rate can consume excessive hard drive space.

- Limit Trender to two instances. This reduces the risk of overrun (out of disk).
- Use WorkstationST Recorders to supplement the use of live Trenders. This allows the user to create unattended data captures that can be triggered by a threshold value or continuously with customizable scan rates.

2.5.1.5 ToolboxST

- Limit ToolboxST to one instance with two concurrent devices opened. This reduces RAM consumption.
- Opening devices on demand allows the user to immediately know from the CMS status icon if they are using the latest revision-controlled copy of the device.

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3 HMI Screen Organization and Design

HMI screens are categorized based on the UX recommendations. A level-based hierarchy is defined and adopted for screen organization within this section. Screen hierarchy is structured to support navigation as well as plant operation. Screen hierarchy by plant topology supports efficiently monitoring the operation and troubleshooting/investigating abnormal situations. In addition, screen content facilitates operator tasks, so that users can efficiently perform their tasks within a limited set of screens. A typical example of a plant control room layout, with respect to HMI screens, is shown below. Many control rooms will have multiple HMI screens.



Control Station Concept

Level 1 screens are designed to be installed within the large overhead monitors to serve as dashboards for plant operation and health status. Level 2 to 4 screens are designed to be displayed within desktop monitors, encouraging the intended user interaction that is discussed later.

3.1 System vs. Process Definition

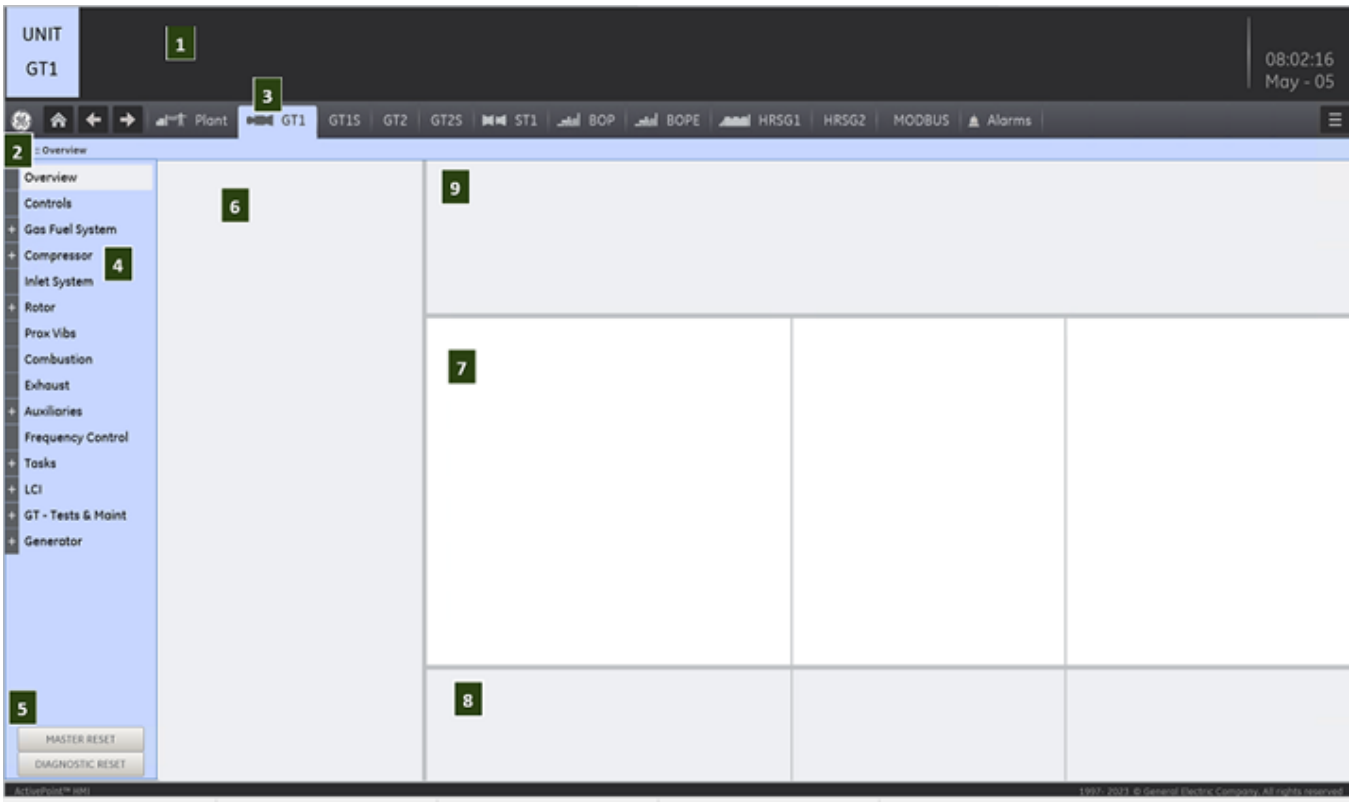
The terms *system* and *process* are important concepts related to HMI screen organization.

System is defined as a set of interacting or interdependent components that form an integrated whole. A *system-centric* screen is one that presents the physical equipment or data pertaining to the physical equipment. For example, in the figures below, Level 3 and Level 4 screens can be considered as a system-centric screen. It shows the valves and pumps that are built and grouped together, representing a system.

Process is defined as a series of actions or steps taken to achieve an end result. In layman's terms, system refers to *what it is*, and process refers to *what it does*. A *process-oriented* screen is one that presents a set of data that is organized to provide the status of the process. It generally contains data from multiple systems. The process screen shows the interaction of multiple systems and allows for adjustments to different systems that all relate to that single process.

3.2 General Screen Template

All screens are developed based on the following Template Screen. Areas 1 to 4 are mandatory and standard for all screens. Areas 5 to 8 can be customized based on screen requirements.



Template Screen (for Level 2 to 4 screens)

Each area of the screen is dedicated to a specific intended use.

1. **Headerbar**: Data that indicates key performance of a plant, device, or sub-system, depending on the level of the screen and the status of key equipment in that level.
2. **Breadcrumbs**: Display the current location with respect to the screen hierarchy so that users can quickly recognize *where they are*.
3. **Tab Navigation Menu**: Used to navigate to Level 2 screens. The Level 2 tab encompasses the side navigation menu, indicating clearly the parent-child structure of the screen hierarchy.
4. **Side Navigation Menu**: Used to navigate to Level 3 and 4 screens.
5. **Functional Buttons**: Additional action buttons that are common to sets of screens. Control buttons that need to be accessed throughout screens under the same parent or buttons that enable users to access different tools or functions can be displayed within this area. However, note that only buttons needed frequently or urgently are placed within this area. Button spacing provides contextual information: Space buttons more closely when they are related.
6. **Controls**: Control buttons or overall status information of a group.
7. **Highlights**: Key information represented more visually (bar charts, trends, and so on).
8. **Details**: Detailed specific data about the group.
9. **Group**: Areas divided and placed based on the process or the physical layout of equipment.

Headerbar

The headerbar of all screens includes the currently selected Block or Unit (if multiple blocks or units are configured), Gross and Net Plant output, Active alarms, Site name, Date, and Time, as shown below.



Headerbar Template

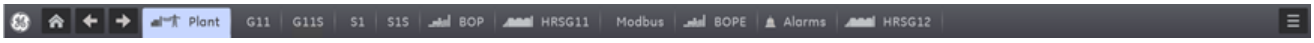
The rest of the area will host key performance indicators (KPIs) for each Level 2 group such as ST, GT, HRSG, or BOP. The following figure shows an example headerbar that is fully designed.



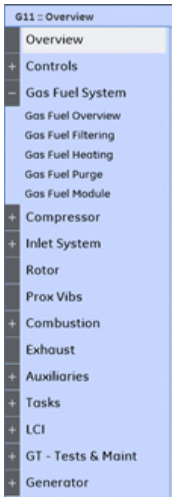
Example: HRSG Headerbar

Navigation Menu Area

All Level 2 to 4 screens include navigation structure unlike Level 1 screens. The navigation structure includes the Tab Navigation Menu and the Side Navigation Menu. The following figures illustrate this navigation structure.



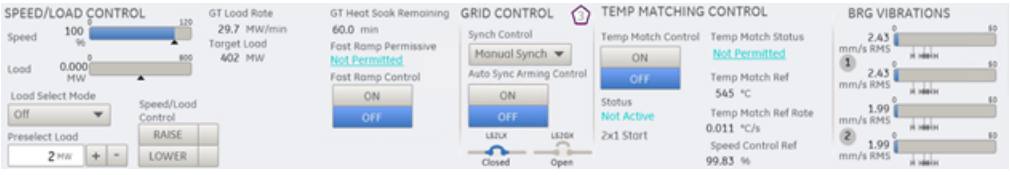
Tab Navigation Menu



Side Navigation Menu

Control Area

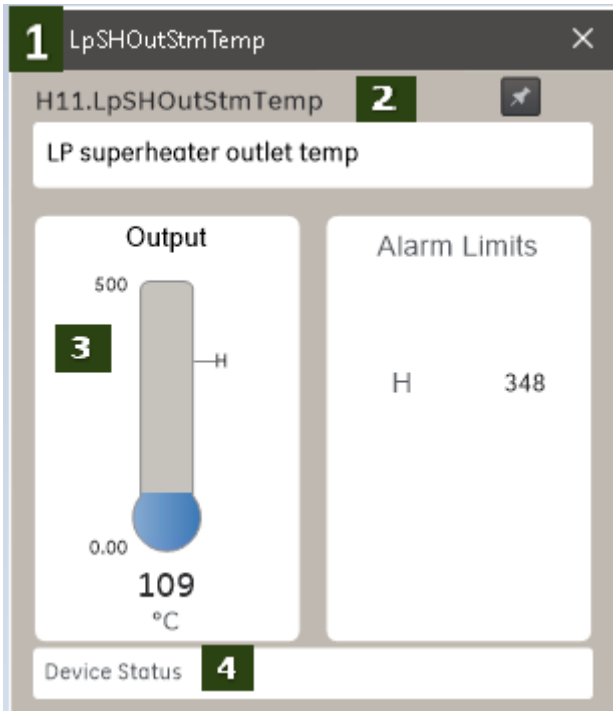
The following figure illustrates an example control area design.



Example: Gas Turbine Control Area

3.3 General Faceplate Template

Faceplates are set up to appear when an interactive element such as a pump or valve is clicked on. Operators launch faceplates whenever they need to take an action to start, stop, or select a state for a device, if drop-down menus are not used. Faceplates are also used to select the mode of operation for a particular device (for example, Auto or Manual). The action of clicking on a mode initiates beginning the event associated with a pump change of state. A new Faceplate design is incorporated to ensure consistency and enhance user interaction. The visual presentation of the Faceplate and the objects associated with it are similar. A sample template for the Faceplate is shown in the following figure.

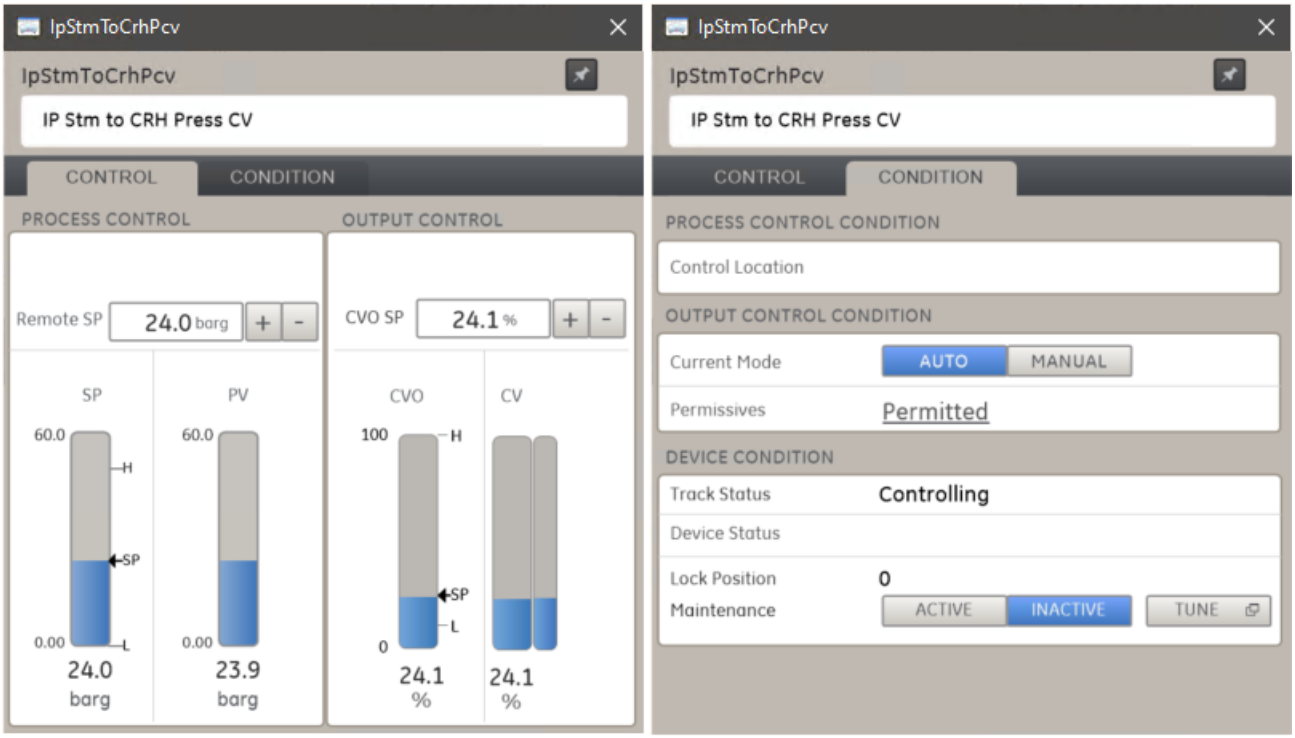


Faceplate Information Layout

Each area of the faceplate is dedicated to a specific intended use.

1. **Title bar**: The description or name of the device related to the faceplate.
2. **Background**: The area to place device information as well as titles of groups included within the faceplate.
3. **Group**: This area is divided based on the characteristics of information and control actions available within the device. Area 3 can vary based on Area 4.
4. **Functional**: This area contains relevant buttons for expanding the details of the device, if available.

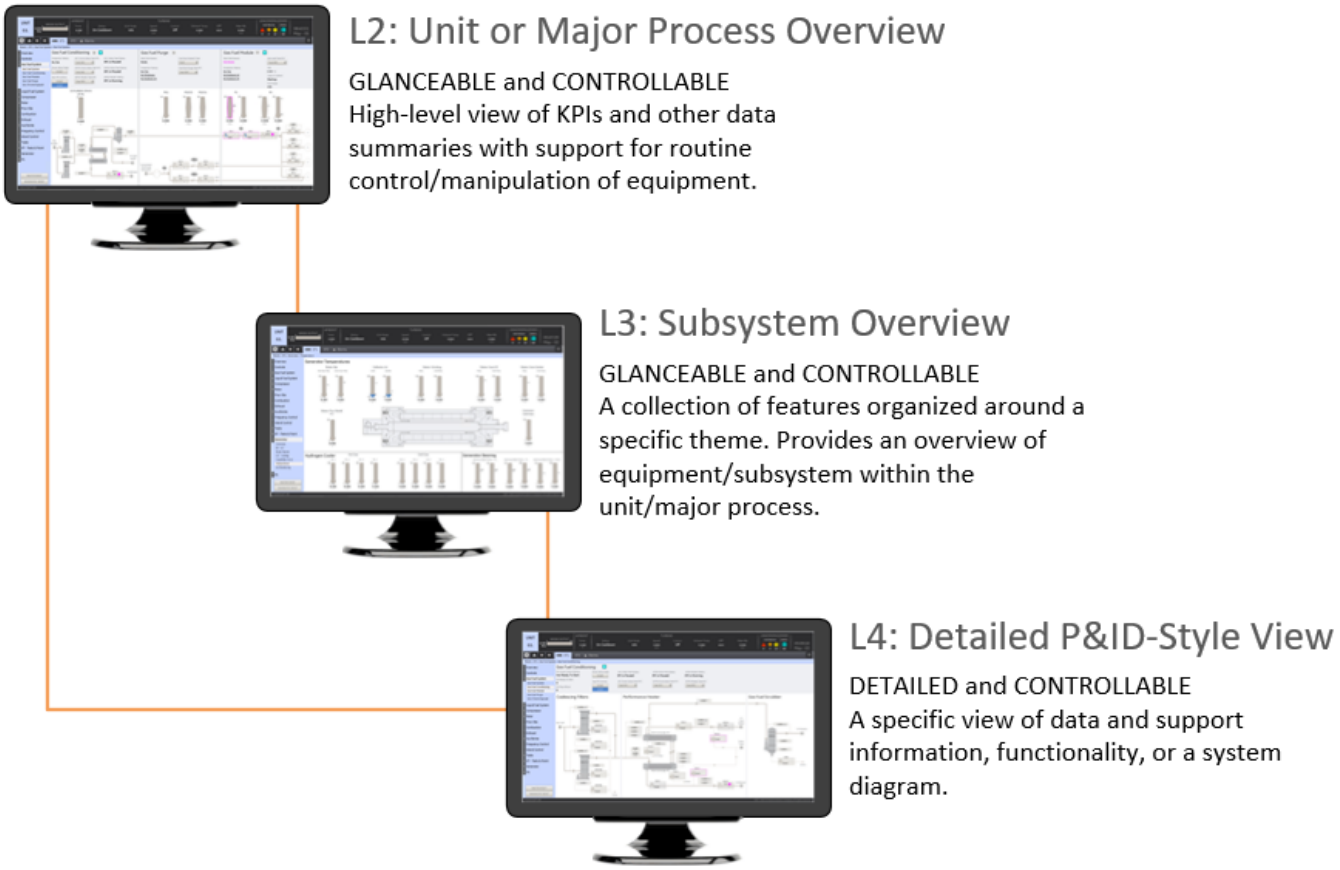
The following figure shows an example faceplate for a Control Valve device or an object within the HMI.



Example: Control Valve Faceplate

3.4 Level Based Screen Hierarchy

All screens are structured based on a four-level hierarchy, a system of layered screens that corresponds to the level of detail and functionality contained within the screen. As shown in *HMI Screen Organization and Design*, Level 1 screens are displayed on the wall, and the Level 2 to 4 screens are displayed within the desktop monitors for operator interaction. Increased depth equates to increased detail and functionality. This layered approach helps users consume and interact with information at a level that is appropriate for their needs. The flattened hierarchy presented in the global navigation enables quick movement. The following figure illustrates the details of the Level 2, 3, and 4 hierarchy in which operator interactions occur.



Screen Hierarchy for Levels 2 to 4

3.5 Level Based Interaction Design Guidelines

Each level is intended for a different use. Level 1 covers a large span of oversight with a carefully selected data set that can represent major processes. As the levels go from 2 to 4, the span of control gets narrower and the level of detailed information gets finer. However, each level incorporates both system-centric and process-oriented elements. The specific characteristics of each level are shown below. The following sub-sections describe each level in more detail with an example screen.

Characteristics of Each Screen Level

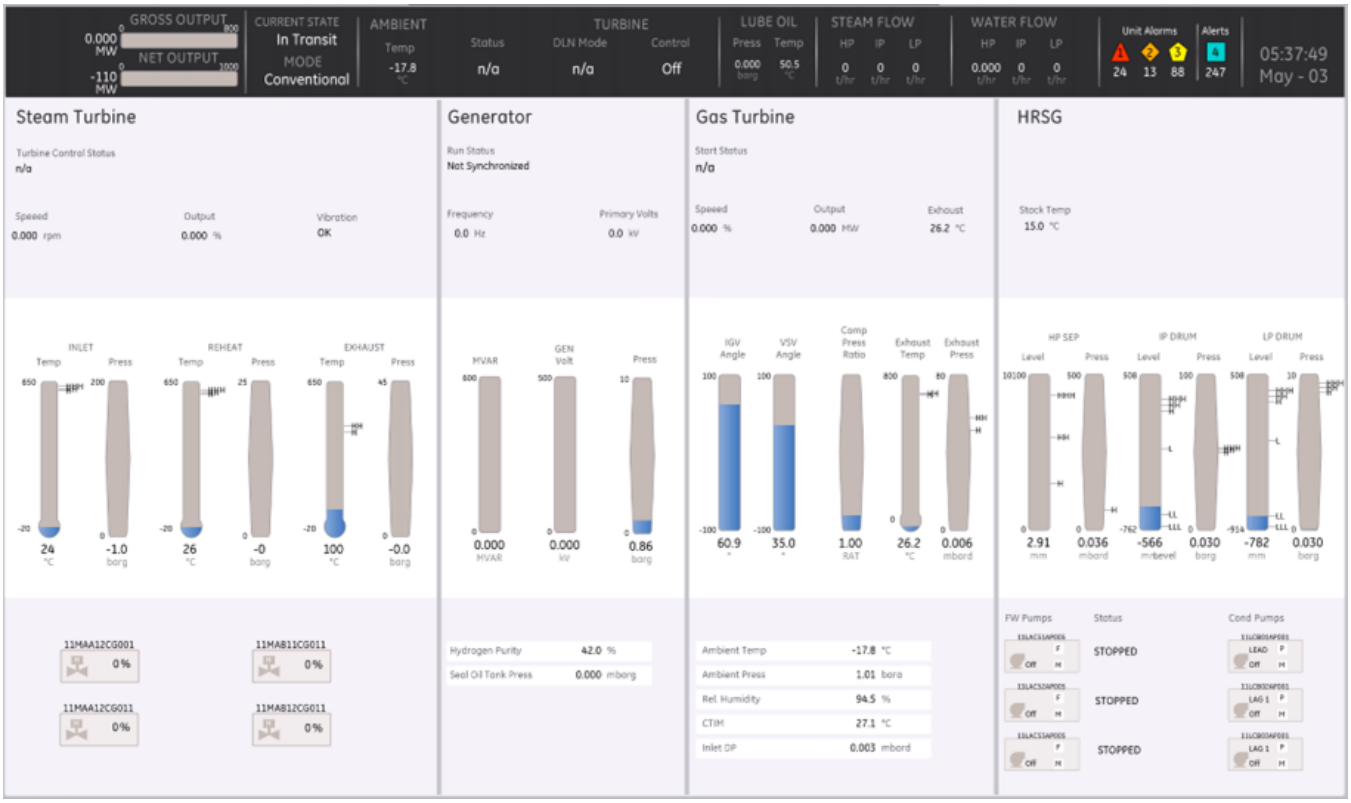
Level	Screen Type	Operator View	Information	Operator Control
1	Process-oriented	Entire power plant	KPIs for plant operations and most critical information that impacts the entire process.	None possible. This is to CONSUME information only.
2	Mostly process-oriented with limited system-centric	Major unit or process inside the plant	Information that: • the operator can do something about. • is required for operators to perform any control functions included in the screen. • the operator needs to recognize quickly when something goes wrong.	• Control functions that operators use frequently. • System items added for major equipment (controlled or not) when the status/diagnostics of that equipment significantly impacts a process. • Selection of those systems' items depends on how the plant is configured.
3	Mix of process-oriented and system-centric	Limited to a sub-component of a system or process	Information that: • the operator can do something about. • is required for operators to perform any control functions included in the screen.	Control actions for rare use only (for example, transferring fuel, swapping certain pumps).
4	Mostly system-centric contents with limited process-initiating controls/content.	The narrowest span of view or control	• P&ID-view style that reflects the process relationship among equipment and components in an abstracted level. • Information that the operator can do something about and information that is required for operators to perform any control functions are included in the screen.	• Controls actions are included when monitoring all equipment during actions. • Access to relevant actual P&ID to the abstract P&ID view if existing.

3.5.1 Level 1 Screens

The Level 1 screen (Plant Overview) is a process area overview that covers the operator's entire span of control. Operators are able to tell, at a glance, the overall operational condition of the plant. Level 1 screens are designed so that all components are compartmentalized for the plant/system. For example, the plant overview screen has four compartments, namely, Steam Turbine, Generator, Gas Turbine, and HRSG. All Level 1 screens have a Headerbar to show plant related parameters. The Level 1 screen is typically hosted on a large wall monitor within the Control Room. Level 1 screens are not designed for operator interaction; however, because they provide overview data on plant conditions, they are considered to be monitoring screens. This screen is launched from the HMI and is part of the HMI screen hierarchy.

Level 1 Screen Characteristics

Type of screen	Process-oriented
Span of operator view/control	Entire power plant, covering the operator's span of control. Provides “at a glance” overall operational condition.
Type of information presented	KPIs for plant operations and most critical information that impacts the whole process of operation.
Operator control	None possible. This screen is only to enable information to be consumed.
Intended user interaction	Operators use this screen for a quick understanding of plant status and any abnormalities.



Level 1 Screen Example: Plant Overview

3.5.2 Level 2 Screens

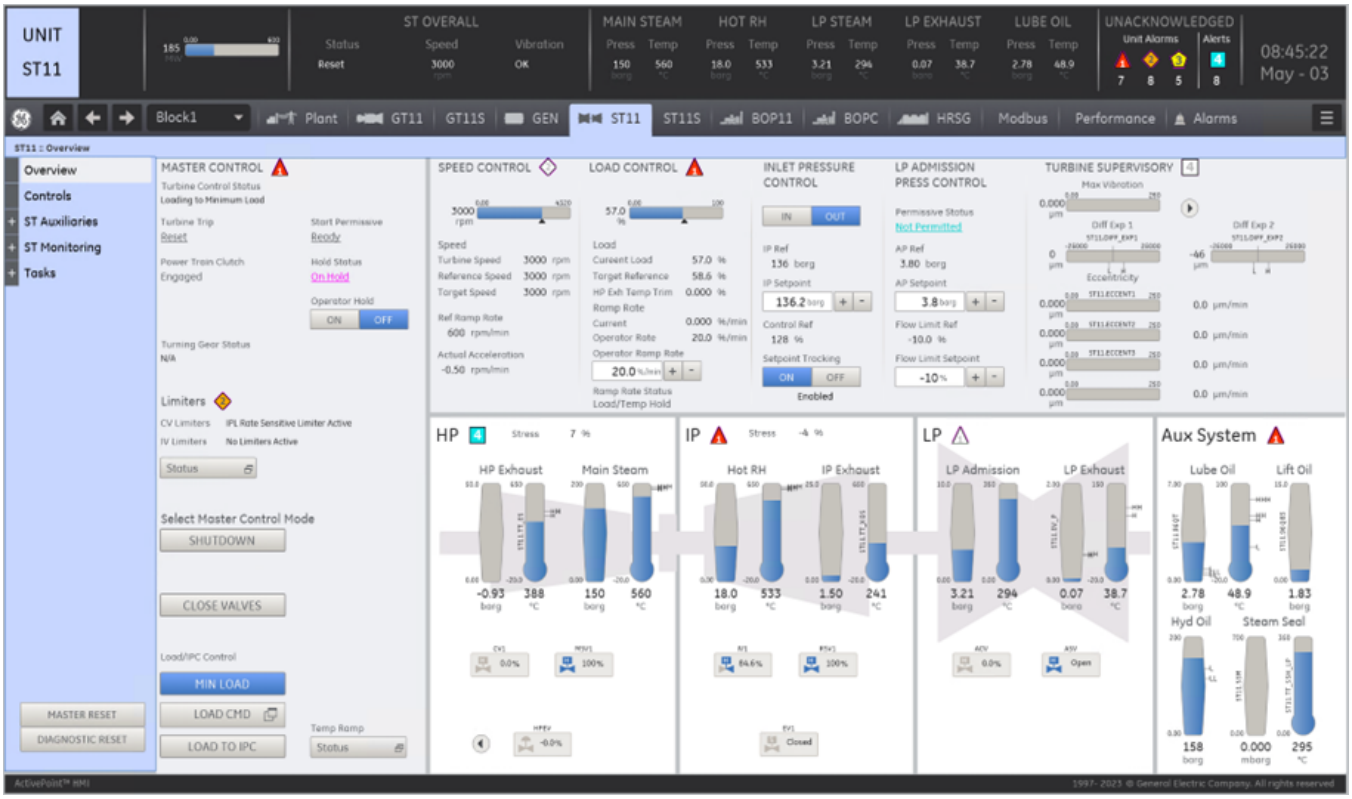
Level 2 screens show the system or major piece of equipment for which the operator is responsible. There is a Level 2 screen for each major system or piece of equipment, such as HRSG, ST, GT, and so on. The overview screen contains all critical KPIs for the system that is required for controlling and monitoring. For example, within the GT Overview screen shown in the following figure, the operator can start, stop, and load the turbine. All Level 2 screens comprise groups of information based on their corresponding sub-systems. Each Level 2 screen has a system-wide Headerbar. The Side Navigation Menu shows all Level 3 and 4 screens for the same system.

Level 2 Screen Characteristics

Type of screen	Mostly process-oriented with limited system-centric
Span of operator view/control	Major unit or process inside the plant
Type of information presented	• Information that the operator can do something about. • Information that is required for operators to perform any control functions is included in the screen. • Information that the operator needs to recognize quickly when something goes wrong.
Operator control	• Control functions that operators use frequently. • System items added for major equipment (controlled or not) when the status/diagnostics of that equipment significantly impacts a process. • Selection of those system items (for example, major equipment within a process) depends on how the plant is configured.
Intended user interaction	• Operators use Level 2 screens most of the time (typically 70– 80%). • Users mainly monitor operation via these screens. • Operators perform routine control tasks via these screens and faceplates that are accessible from the screens.



Level 2 Screen Example: Gas Turbine Overview



Level 2 Screen Example: Steam Turbine Overview



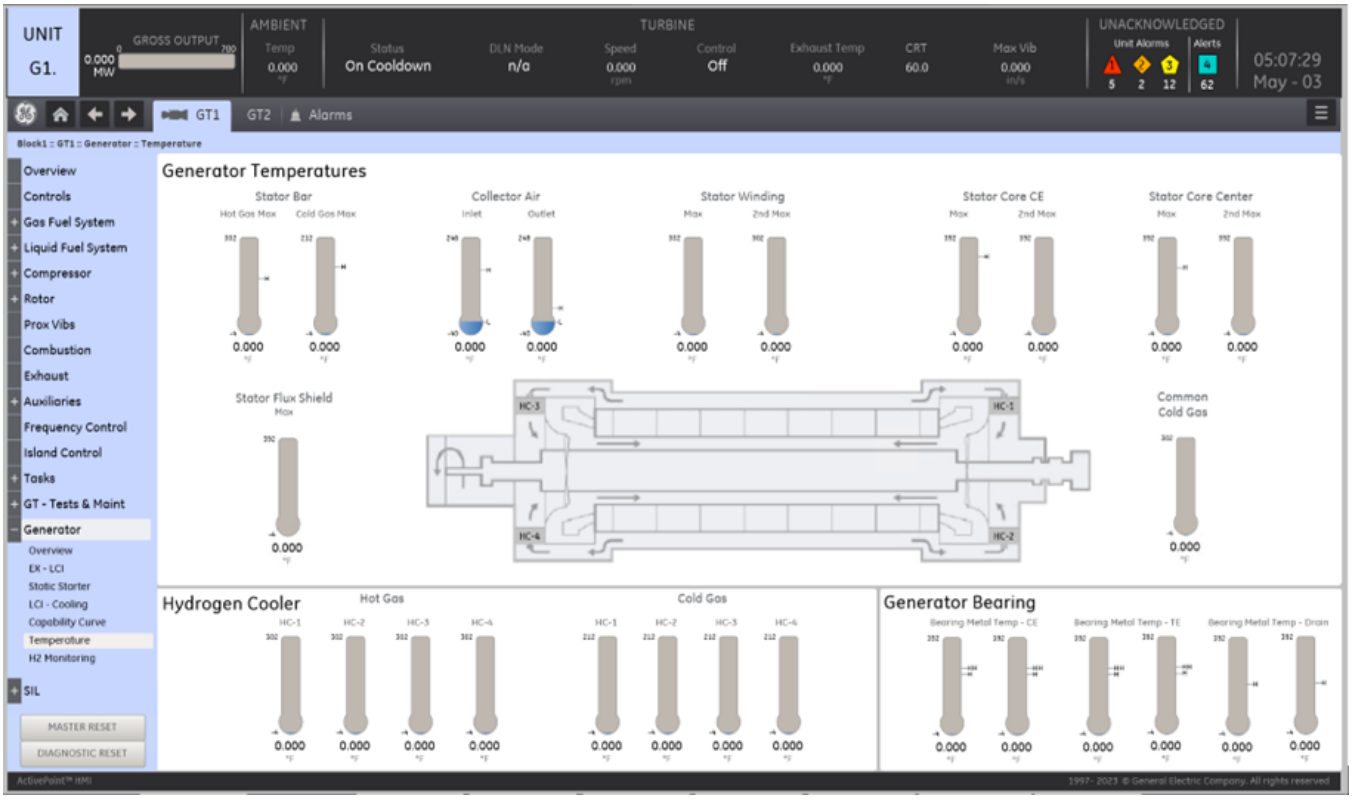
Level 2 Screen Example: HRSG Overview

3.5.3 Level 3 Screens

Level 3 screens are the overview screens for a subsystem related to Level 2 components. Level 3 screens provide details about a single sub-system, piece of equipment, or process control scheme. Level 3 screens are used to display and control specific and detailed information related to the sub-system. For example, the ST Monitoring system screen shown in the following figure is a Level 3 screen related to the monitoring of critical Steam Turbine temperatures and vibrations. The screen is also compartmentalized based on multiple sub-systems such as HP, IP, and LP sections. All information that illustrates the overall status of each system is presented within this screen.

Level 3 Screen Characteristics

Type of screen	Mix of process-oriented and system-centric
Span of operator view/control	Limited to a sub-component of a unit or a process
Type of information presented	• Information that the operator can do something about. • Information that is required for operators to perform any control functions included in the screen.
Operator control	Control actions for random or rarely used functions only (for example, transferring fuels and swapping certain pumps)
Intended user interaction	• Operators use this screen when a particular sub-process or sub-unit requires focused attention. • Operators perform routine control tasks via these screens and Faceplates that are accessible from the screens.



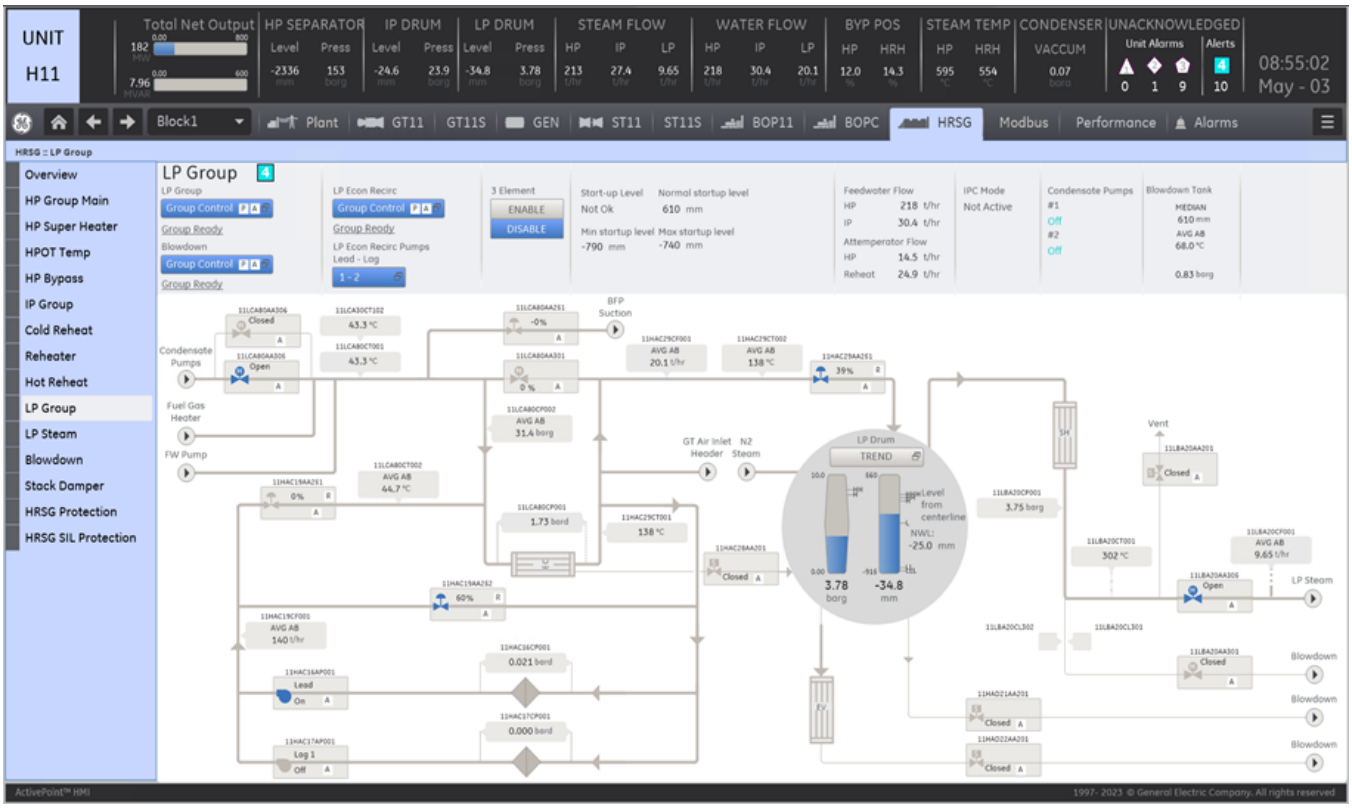
Level 3 Screen Example: Generator Temperatures

3.5.4 Level 4 Screens

Level 4 screens are equipment support and diagnostic displays, showing items such as interlocks, emergency shutdown test screens, and diagnostic screens. These screens are generally Faceplates or P&IDs with support and diagnostic information. When using P&ID, the layout is consistent with a process flow (for example, directed from left to right and from top to bottom).

Level 4 Screen Characteristics

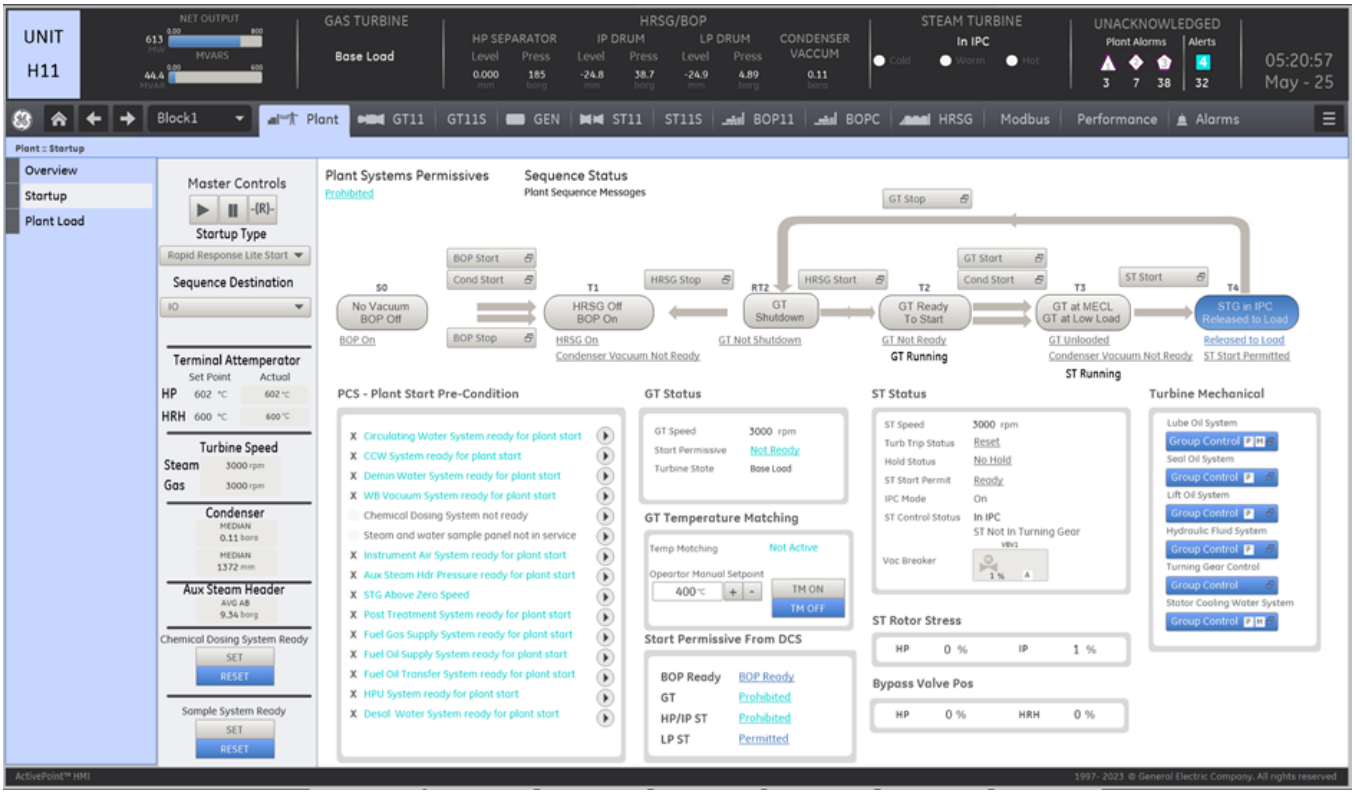
Type of screen	Mostly system-centric with limited process-initiating controls/contents and component-level details
Span of operator view/control	The narrowest span of view or control
Type of information presented	• P&ID-view style that reflects the process relationship among equipment and components in an abstracted level. • Information that the operator can do something about and information that is required for operators to perform any control functions are included in the screen.
Operator control	• Controls actions are included when monitoring all equipment during actions. • Access to the relevant actual P&ID to the abstract P&ID view, if existing.
Intended user interaction	• Operators use these screens to troubleshoot, diagnose, and resolve any abnormal situation. • Operators use these screens to check, confirm, and verify the most detailed level of information. • Operators perform specific troubleshooting tasks via these screens and Faceplates that are accessible from the screens.



Level 4 Screen Example: HRSG LP Group

3.6 Task Based Screens

Tasks like plant startup, shutdown, and so on are incorporated into the Side Navigation Menu under the associated main system tab (for example, GT, ST, or Generator) or the Plant tab so that the operator can directly view these screens to enable these actions. An example is provided in the following figure. Tasks are Level 3 screens. If there is a sub-level required for any of screen, a Level 4 screen can be created.



Task Based Screen Example: Plant Startup

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4 Alarm Management Philosophy

Alarm Rationalization is the cornerstone of GE Vernova's Alarm Management Philosophy. The purpose of the rationalization process is to reduce the actionable alarms across the plant and improve operator effectiveness by offering substantial guidance and prioritization of items that require operator intervention to avoid consequence. Following the process also documents common alarm philosophy, definitions, rationale, and prioritization rules, including the display philosophy.

GE Vernova's Alarm Management Philosophy is based on ISA 18.2 standard guidelines for alarm management. It has been adapted for application across GE Vernova's GT, ST, Generator, HRSG, and BOP. There are two main activities that are critical to the reduction of alarms that are presented in a contextual manner to an operator for timely action.

- Reduction of configured alarms that are presented to the operator (engineering activity).
- Tuning instruments, set points, and dead bands based upon actual equipment response times to reduce spurious alarms (commissioning activity).

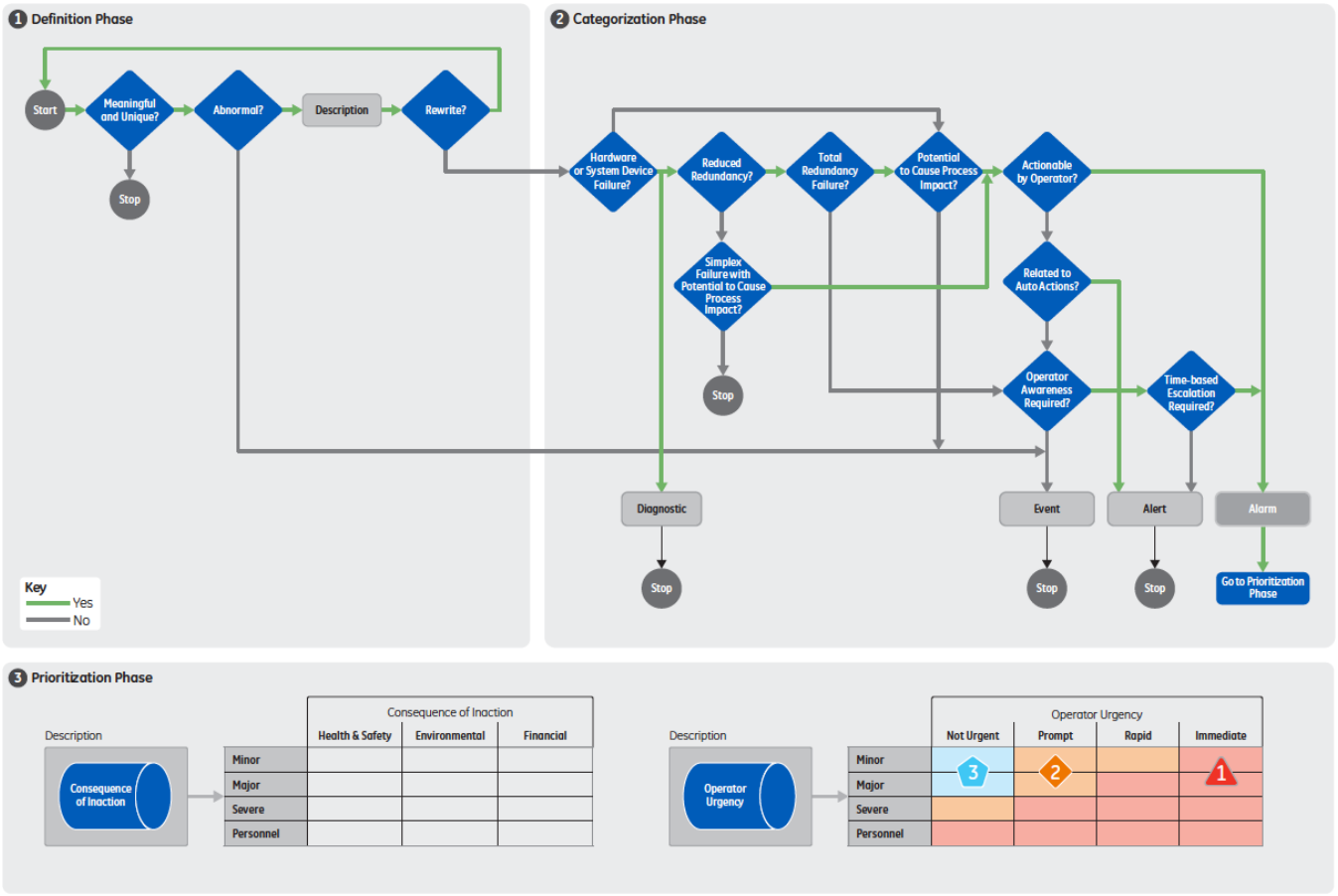
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5 Alarm Rationalization

This section describes the overall alarm rationalization process and each step within it.

5.1 Process

Plant alarms are an aggregate of component parts and systems; therefore, they require common philosophies and rationalization rules across all equipment within the plant. This document provides an overview of the three-step process (Definition, Categorization, and Alarm Prioritization). It requires design teams to be fully conversant with the process and understand respective responsibilities to bring value and operator effectiveness by reduction of plant alarms. The steps are designed to be thought-provoking and to drive rationale for categorizing potential alarms as Events, Diagnostics, Alerts, or Alarms. The following figure depicts the Alarm Rationalization Process.



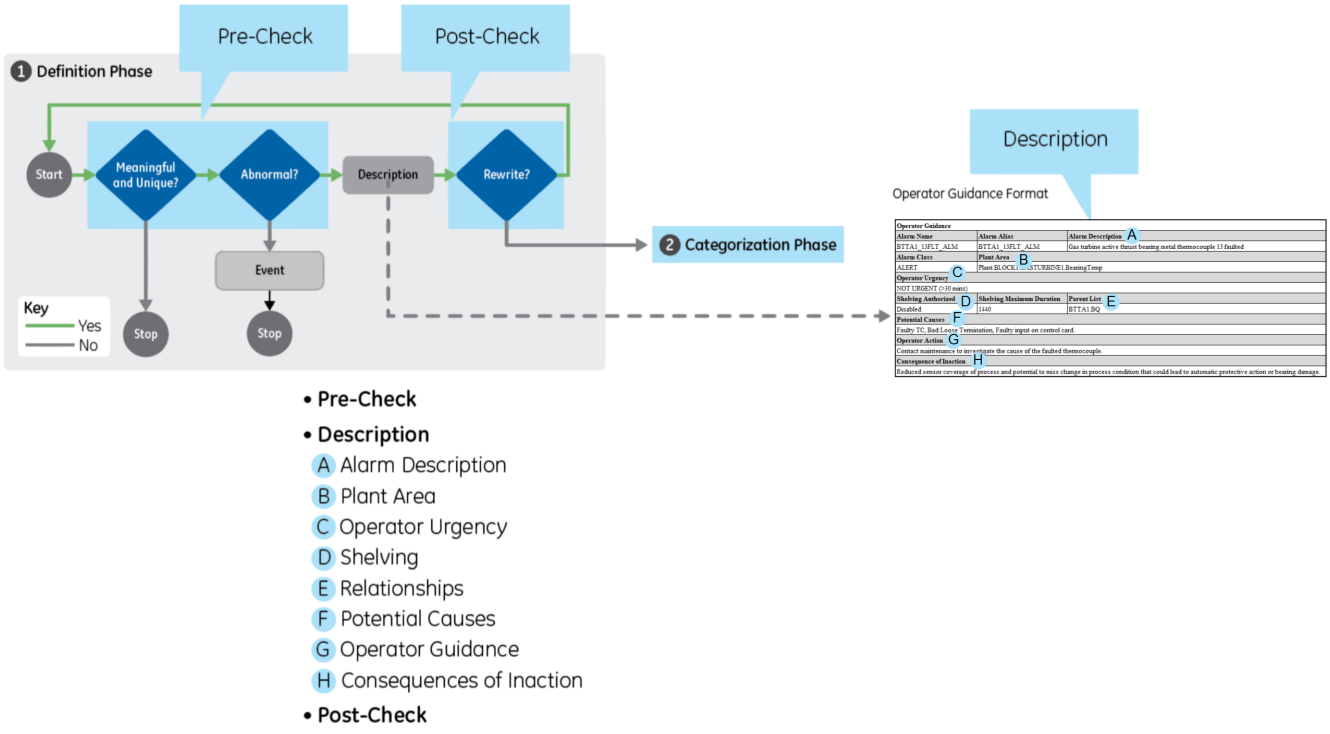
Alarm Rationalization Process

To fully understand the Alarm Rationalization Process and Alarm Management Philosophy, one must understand the annunciation display philosophy and interaction strategy for operators and other plant personnel.

- Primarily, operators take actions on a priority basis to avoid process impact.
- The **live Alarm Display** presents to operators for their action alarms with a priority that includes the urgency.
- The **live Alert Display** presents alerts to operators that require their awareness but are not directly actionable by them. These items may require the operator to communicate and collaborate with other personas to correct issues (such as maintenance).
- The **live Diagnostics Display** and **Historical Log Display** (Alarms, Alerts, Diagnostics, and Events) present items to maintenance personnel or the plant engineer for diagnoses of equipment failures and issues.
- Alarms and Alerts must be acknowledged by an operator.
- Diagnostics and Events are typically acknowledged by maintenance personnel.

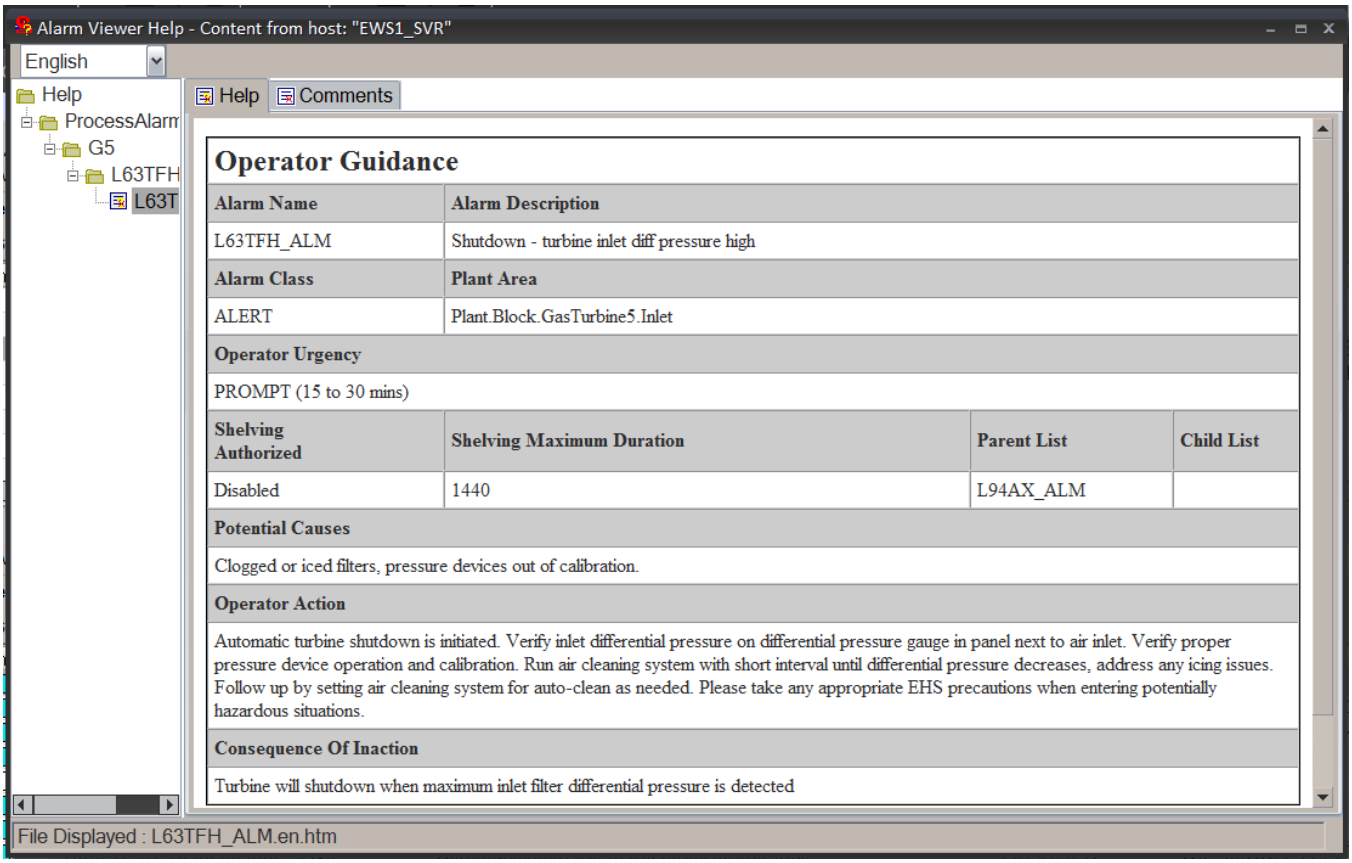
5.2 Definition Phase

The definition process challenges the rationale behind inclusion of potential alarms as actual alarms, alerts, or diagnostics. It requires component, system, and adjacent system-level expertise to evaluate the design parameters of the potential alarm. This phase consists of the following six steps as depicted in the following figure.



Definition Phase of the Alarm Rationalization Process

1. Pre-Check of all potential alarms within a system
   Pre-checks are designed to determine if a potential alarm meets the criteria of being an alert or alarm or whether re-coding is necessary. After any iteration of re-coding is completed, the potential alarm is then re-evaluated to determine if it is meaningful, unique, and abnormal.
2. Description of the potential alarm
   This step documents the variable name, its description, and any design documentation reference used while rationalizing the alarm. It also documents the conditions under which the alarm is active, whether the operator is allowed to shelve it, and the plant area in which the alarm belongs.
3. Relationship of potential alarms within a given system
   The relationship between potential alarms within a system is explored to determine if any highly managed alarm techniques, such as parent/child relationships, can be applied.
4. Relationship of potential alarms to alarms in other/adjacent systems
   The relationship between potential alarms and alarms within other/adjacent systems is explored to determine if any highly managed alarm techniques, such as parent/child relationships, can be applied.
5. Operator guidance creation
   This step documents potential causes, consequences of inaction, operator action, and operator urgency to form an Alarm Help File (shown below) that is accessible to the operator to help the operator decide which actions need to be taken. This step tends to be very iterative.

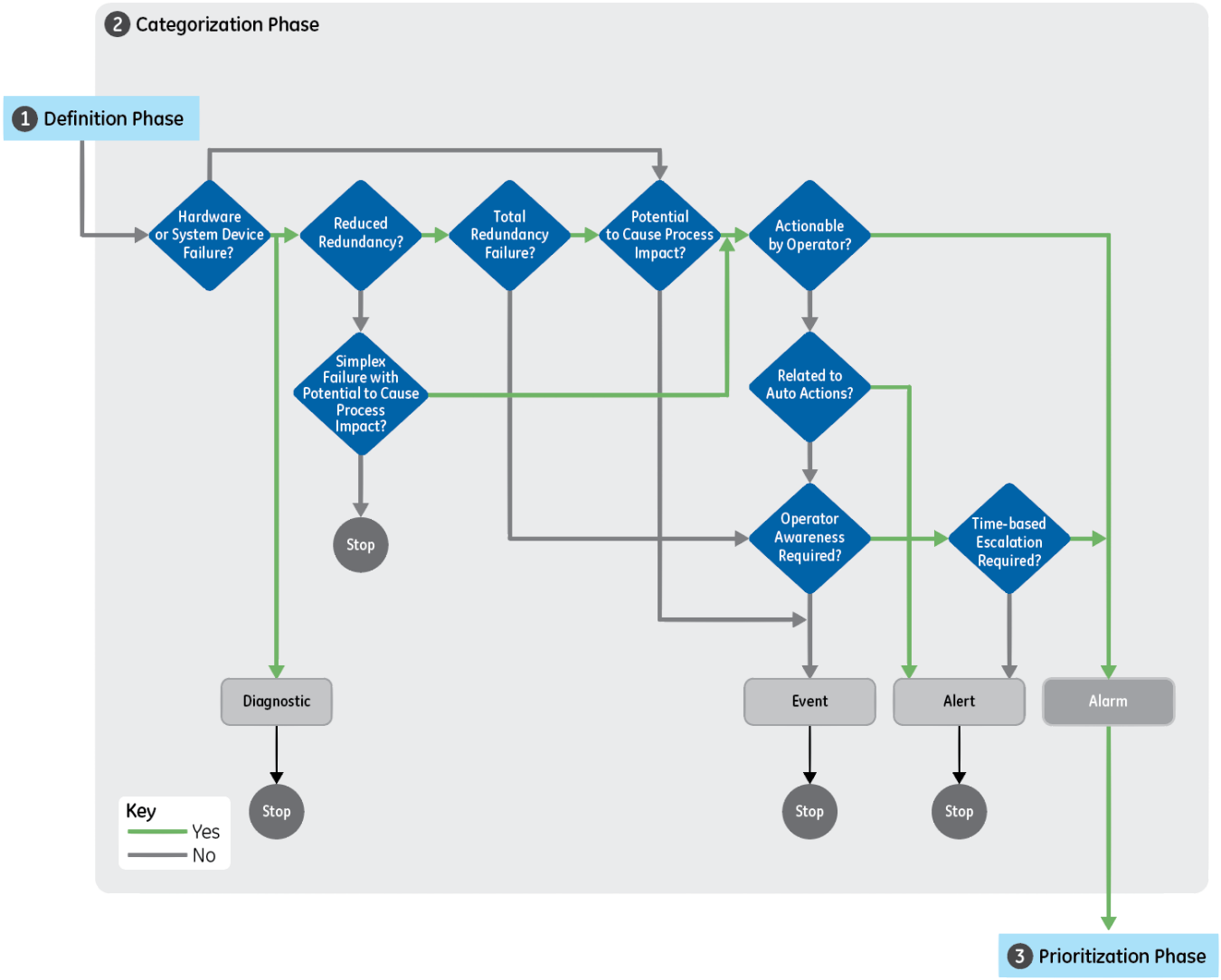


Example: Alarm Help File

6. Post-Check for code re-writing
   It is expected that re-writing of code is required to support rationalization of potential alarms. If so, all steps need to be repeated for the re-coded potential alarm.

5.3 Categorization Phase

Once a potential alarm has gone through the definition phase, the next phase is categorization or the process of separating alarms into annunciation types (Event, Diagnostic, Alert, or Alarm) based on common requirements. The Categorization Phase is detailed in the following figure.

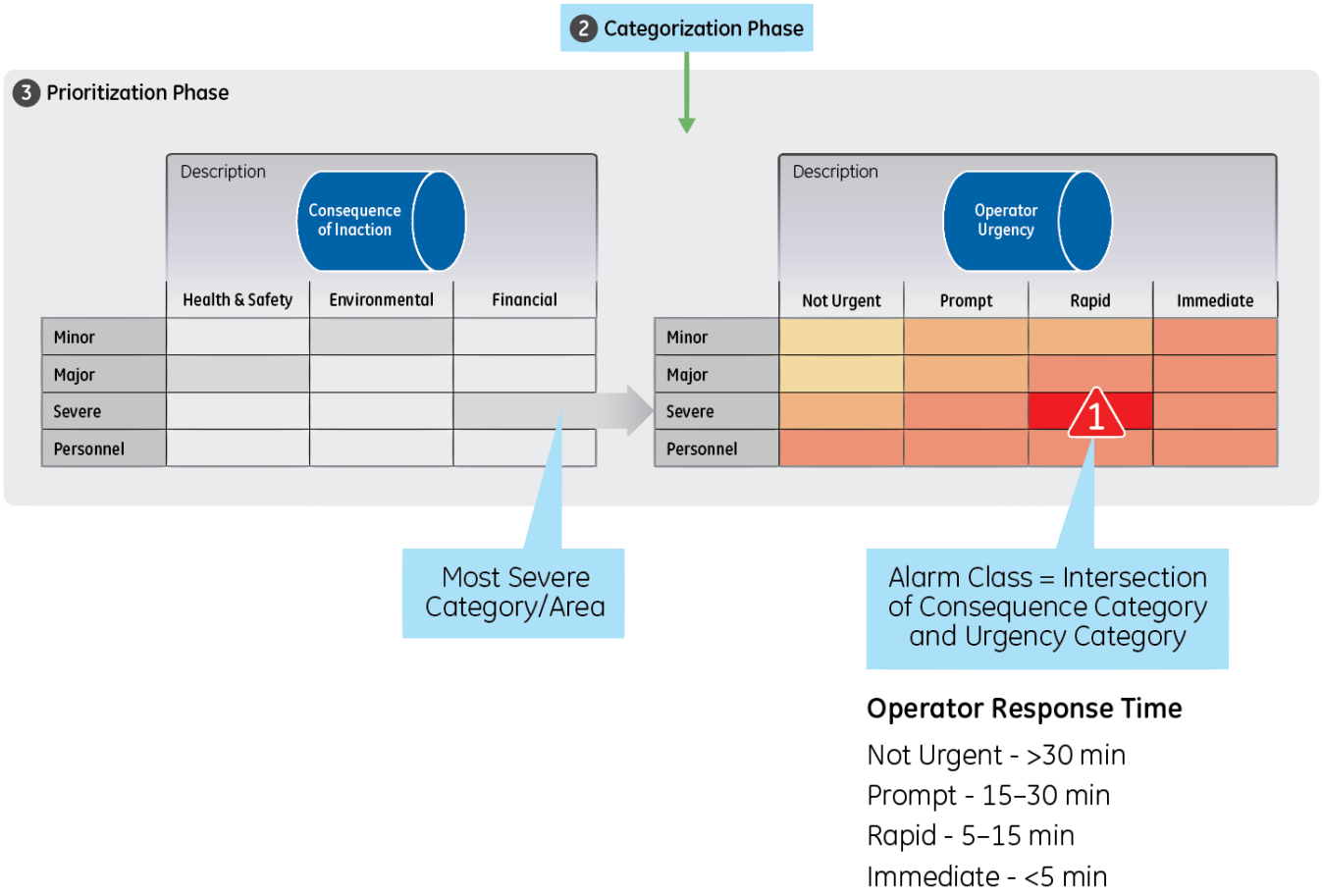


Categorization Phase of the Alarm Rationalization Process

Alarm Annunciation Type	Indicates an abnormality that is meaningful, unique, has the potential to cause process impact, and is actionable by operator within allowable response time.
Alert Annunciation Type	Indicates an abnormality that is meaningful, unique, has the potential to cause process impact, and is NOT actionable by operator but requires the operator to be aware of the situation.
Diagnostic Annunciation Type	Indicates a hardware or software failure (sensor, control device, controlled device, network, or HMI) or communication error with a field device or component. This type includes all smart device warnings or failure annunciations (for example, Foundation Fieldbus device alerts).
Event Annunciation Type	Indicates a state or condition that is meaningful, unique, and NOT abnormal but allows for ease of troubleshooting in understanding order of operation after an excursion (for example, Valve open/close or Breaker open/close).

5.4 Prioritization Phase

Once a potential alarm has been categorized as the Alarm Annunciation Type, it then must be prioritized, which is the process of assigning a level of operational importance to an alarm. This is accomplished by assigning an Alarm Class, which indicates the urgency of response (seriousness of consequences and allowable response time). The following figure depicts the prioritization process.



Alarm Prioritization Phase of the Alarm Rationalization Process

Using the Consequence of Inaction determined during the Definition Phase, a Consequence Category (Minor, Major, Severe, and Personnel) is determined for Health and Safety, Environmental, and Financial aspects. The most severe Consequence Category among these three aspects is used as the Consequence of Inaction severity. The intersection of the Operator Urgency determined during the Definition Phase and the Consequence of Inaction severity within the matrix in the above figure is used to determine the associated Alarm Class.



Public Information